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Graduate QC Analyst · Analytical Scientist · Microbiologist

MSc Pharmaceutical Biotechnology - Merit, De Montfort University
BPharm (First Class Equivalent) - Dr. MGR University, India

RP-HPLC | ATR-FTIR | UV-Vis Spectroscopy | GPC Purification | ICH Stability Studies | GMP Documentation | Aseptic Technique

QC ANALYST · PHARMACEUTICAL BIOTECHNOLOGY · ANALYTICAL SCIENCE

I am a pharmaceutical biotechnology graduate with hands-on experience in analytical chemistry, microbiology, and quality-focused laboratory work. My MSc research involved the synthesis and characterisation of a fluorescently labelled insulin conjugate using RP-HPLC, ATR-FTIR, and UV spectroscopy, work that produced a **96.2% purity** result and **zero contamination incidents** across the entire study. I am looking for a QC Analyst or Analytical Scientist role in UK pharmaceutical manufacturing or a CRO where I can build on that foundation and contribute from day one.

CORE COMPETENCIES

Core Technical Competencies

Instrumentation RP-HPLC (Shimadzu Prominence; LabSolutions software), ATR-FTIR (Bruker), UV-Vis Spectroscopy, GPC / Sephadex G-25 Purification, Lyophilisation	**Quality & Compliance** GMP documentation, SOP writing & review, audit-ready record-keeping, LIMS workflows (self-study in progress), data integrity principles
Analytical Methods Method development & optimisation, ICH-aligned stability studies, assay development, OOS & deviation identification, root cause analysis / CAPA	**Software & Data** Shimadzu LabSolutions (HPLC data processing), Microsoft Excel (data analysis), ChemSketch, standard statistical analysis
Microbiology Aseptic technique, Gram staining, agar plating, broth culture, colony counting, mammalian cell culture (HEK293T, Class II BSC)	**Additional Science** Pharmaceutical quality testing (dissolution, hardness, friability), DNA extraction, gel electrophoresis, PK/PD principles

ABOUT ME

About Me

I grew up in Tamil Nadu, India, and completed my Bachelor of Pharmacy at SRIPMS under Dr. MGR University, graduating with a first-class equivalent result. The course gave me my first real laboratory experience like titrations, dissolution testing, tablet hardness analysis, gram staining, agar plating, broth culture, and aseptic technique. More than the techniques themselves, it gave me an understanding of why precision matters in a pharmaceutical setting. A small error in a dissolution test or a contaminated culture plate is not just a failed experiment. It represents a gap between what a medicine claims to do and what it actually does in a patient.

That understanding is what pushed me to go further. I moved to the United Kingdom in 2023 to study for an MSc in Pharmaceutical Biotechnology at De Montfort University in Leicester. The academic structure was completely different from what I had known in India, more independent, more research-driven. It took adjustment, but I committed to it fully. Over the course of the year, I trained on instruments I had only read about: a Shimadzu Prominence HPLC system, a Bruker ATR-FTIR spectrometer, UV-Vis spectroscopy equipment, and GPC columns for purification. I also spent a week working with mammalian cell culture in a Class II Biological Safety Cabinet, an experience that sharpened my aseptic discipline significantly. I completed the MSc with a Merit.

Since graduating I have been working as a Warehouse Team Lead at Royal Mail while I continue my job search. On the surface that sounds unrelated, but the role has reinforced skills that translate directly to a QC environment: enforcing standard operating procedures, training team members, maintaining zero-error accuracy under sustained pressure, and keeping documentation audit-ready at all times. I was promoted into the role based on my track record, and my team holds the site record for the highest number of parcels sorted in a single shift, 25,000 in nine and a half hours, with full accuracy. The mindset that produces that result is the same mindset I will bring to a QC laboratory.

I want to work in quality control because the stakes are genuinely high. Every batch certificate I sign off, every OOS investigation I contribute to, every contamination I prevent, those actions connect directly to patients who depend on the medicines being right. That is not a comfortable thought. It is a motivating one. I work best when something real depends on getting it right, and I have spent the last several years proving that to myself in different settings. I am ready to prove it in a laboratory.

EDUCATION

Academic Background

Qualification	Institution	Year	Result
MSc Pharmaceutical Biotechnology	De Montfort University, Leicester	2023 – 2024	Merit
Bachelor of Pharmacy	SRIPMS, Coimbatore (Dr. MGR University)	2018 – 2022	First Class Equivalent

MSC RESEARCH PROJECT

Fluorescent Labelling of Insulin with EITC

Fluorescent Labelling of Insulin (Actrapid) with Eosin Isothiocyanate: A Comprehensive Analysis

De Montfort University, Leicester · September 2023 to September 2024 · Individual research project · Outcome: Merit

Background and Aim

Fluorescently labelled insulin is used in diabetes research to track how insulin moves through cells and tissue, it allows scientists to study pharmacokinetics and receptor binding without relying on radioactive tracers. The challenge is producing a mono-labelled conjugate, meaning exactly one fluorescent dye molecule attached to one specific site on the

insulin molecule. If too many dye molecules attach (di- or tri-labelled), the conjugate behaves differently from native insulin and the analytical data becomes unreliable.

Eosin Isothiocyanate (EITC) was selected as the fluorescent label over the more commonly used FITC, because EITC offers significantly better photostability, it does not bleach as quickly under excitation light, which is important for any extended imaging or spectroscopic work. The aim of the project was to synthesise, purify, characterise, and assess the stability of EITC-labelled Actrapid insulin under systematically varied reaction conditions.

Experimental Methods

Synthesis: Actrapid (commercial human insulin, 100 IU/mL) was mixed with EITC stock solution (25 mg/mL in acetone) at two molar ratios: 1.5:1 and 2:1. The reaction pH was adjusted to 7.0 using 0.1M NaOH and HCl. The mixture was stirred slowly in the dark at room temperature across reaction times of 18, 20, 24, 26, and 28 hours.

GPC Purification: After each reaction, Gel Permeation Chromatography using a Sephadex G-25 column (Buchii B688 system) separated the EITC-Insulin conjugate from unreacted free dye and unlabelled insulin. Fractions were collected in phosphate buffer at pH 7.4 and visually assessed for band resolution before proceeding.

Lyophilisation: Collected conjugate fractions were first frozen using liquid nitrogen and then lyophilised overnight to produce a dry powder. The powder was stored at -20°C to preserve stability ahead of HPLC analysis.

RP-HPLC Analysis: A Shimadzu Prominence system was used with a Phenomenex Luna C18(2) column (150 x 4.6mm, 3 micron). Gradient elution used 0.1% TFA in water and 90% acetonitrile with 0.1% TFA. A fluorescence detector (excitation 521nm, emission 544nm) identified labelled conjugates, while a PDA detector at 276nm tracked unlabelled insulin. Each reaction condition was analysed to build a full optimisation dataset.

ATR-FTIR Characterisation: A Bruker FTIR spectrometer scanned Actrapid (liquid and air-dried), pure EITC powder, and the final conjugate across 4000 to 400 cm-1. The goal was to confirm whether conjugation had occurred and whether the structure of insulin had been preserved.

14-Day ICH Stability Study: The conjugate was dissolved in two separate media: distilled water and Actrapid diluting fluid and stored at 24°C in a light-restricted environment. UV absorbance was recorded daily across 300 to 600nm over 14 days to monitor any degradation, aligned to ICH stability testing principles.

Experimental Workflow

① Synthesis	② GPC Purification	③ Lyophilisation	④ RP-HPLC	⑤ ATR-FTIR	⑥ UV Stability
EITC + Actrapid	Sephadex G-25	Liquid N_2 → powder	Shimadzu / LabSol.	Bruker	14-Day ICH

MSC RESEARCH — RESULTS

Research Results & Key Findings

The HPLC data showed a clear relationship between reaction time and product purity. At 18 hours, the chromatogram produced a single dominant peak at 17.05 minutes representing the mono-labelled B1 conjugate, accounting for **96.2% of the total product**. Extending the reaction to 24 or 28 hours caused additional peaks to appear di- and tri-labelled species reducing the mono-label proportion to 77% and 67% respectively. This confirmed that 18 hours was the optimal reaction time.

Increasing the EITC to insulin molar ratio beyond 1.5:1 did not increase mono-labelling. At a 2:1 ratio, the proportion of multi-labelled species increased, suggesting that once the primary amine binding sites are occupied, excess EITC reacts with secondary sites. Similarly, raising pH above 7.0 increased the nucleophilicity of the amino groups and promoted over-labelling, reducing the mono-label yield from 96% to around 75%.

The ATR-FTIR spectra confirmed successful conjugation. In the free EITC spectrum, a sharp isocyanate peak was visible at 2023 cm-1. In the Insulin-EITC conjugate spectrum, this peak was absent and replaced by an S=C stretching vibration at 2136 cm-1, confirming that a thiourea bond had formed between EITC and the primary amine groups on insulin. The N-H and carbonyl peaks characteristic of insulin were retained, confirming that the protein structure remained intact after labelling.

The 14-day stability study showed that when dissolved in distilled water, the conjugate maintained stable absorbance with only minor variation (1.378 on day 0 to 1.217 on day 14). In the Actrapid diluting fluid, there was an initial decline from

day 0 to day 3, followed by a period of relative stability, with an unexpected increase observed on day 14. This suggested the diluting fluid may interact with the conjugate and that distilled water is the more suitable storage medium.

Summary of Key Findings

Variable Tested	Finding	Relevance to QC Practice
Reaction time	18 hours → 96.2% mono-labelled product	Method development and specification setting
Molar ratio (EITC:Insulin)	2:1 increased multi-labelled by-products	Reagent concentration as a critical process parameter
Reaction pH	pH above 7.0 reduced mono-label yield to ~75%	pH control as a critical quality attribute
FTIR confirmation	Thiourea bond at 2136 cm-1; insulin structure intact	Structural integrity verification post-conjugation
Stability in distilled water (14 days)	Absorbance stable: 1.378 to 1.217	ICH-aligned shelf-life and storage assessment
Contamination events	Zero across the entire project	Aseptic technique and GMP work practice

UNDERGRADUATE LABORATORY EXPERIENCE

Undergraduate Laboratory Experience

Bachelor of Pharmacy - SRIPMS, Coimbatore · Dr. MGR University · 2018 to 2022 · First Class Equivalent

The BPharm programme provided five years of structured laboratory training across analytical chemistry, pharmaceutical quality testing, medicinal chemistry, and microbiology. The work was practical and cumulative; each year built on the last, and by the final year I was comfortable working independently with a range of techniques and equipment.

Analytical chemistry: Titration methods, gravimetric analysis, accurate pipetting, four-decimal-place weighing, and spectrophotometry. Early training in the principle that measurement uncertainty must be understood and controlled.

Pharmaceutical quality testing: Tablet hardness, dissolution, disintegration, and friability testing: the same tests that sit at the core of finished product QC in any manufacturing environment. Working to pharmacopoeia specifications.

Medicinal chemistry: Drug structure-activity relationships, synthesis principles, and a thorough grounding in PK/PD: how drugs are absorbed, distributed, metabolised, and eliminated. This knowledge informs how I think about stability data and API behaviour.

Microbiology: Agar plate preparation, streak plating, Gram staining, broth culture development, colony counting, incubation management, and aseptic room technique. This is where my contamination-prevention mindset was formed.

Molecular biology: DNA extraction, gel electrophoresis, and ligation techniques: foundational methods relevant to bioanalytical and microbiological QC.

MAMMALIAN CELL CULTURE

Mammalian Cell Culture

MSc Module - De Montfort University · One-week intensive practical

Midway through the MSc, I completed a week of mammalian cell culture work using HEK293T cells (Human Embryonic Kidney line). The first day covered cell biology theory, the equipment, and the techniques involved. The following three days were hands-on: seeding cells, monitoring growth overnight, passaging, cell counting and repeating the cycle. All work was carried out in a Class II Biological Safety Cabinet, the cleanroom-equivalent environment for cell culture work.

The culture produced successful cell growth with zero contamination events. I wrote a personalised SOP for the cell passage and maintenance process - documenting each step, the critical control points, and the actions required if

contamination was detected. Writing that SOP was the point where I understood that a good protocol is not just a procedure; it is a record that someone else could follow in your absence and get the same result.

PROFESSIONAL EXPERIENCE

Work Experience

Warehouse Team Lead - Royal Mail

Northampton, United Kingdom · October 2023 to present · Promoted on Merit

I have been working at Royal Mail since arriving in the UK, initially as a warehouse operative and subsequently promoted to Team Lead based on my accuracy record and operational reliability. The role involves managing a team of six, ensuring the daily processing targets are met without compromising accuracy, and maintaining full compliance with health, safety, and documentation requirements.

- Supervised and trained six operatives in SOP adherence, safe working practices, and quality verification procedures. New team members are signed off only when they can demonstrate the required standard consistently.
- Managed throughput of 20,000 to 25,000 items per shift while maintaining 99% documented accuracy. The team holds the site record: 25,000 parcels sorted in a nine-and-a-half-hour shift including a one-hour break.
- Maintained zero recorded sorting errors across two consecutive reporting cycles by applying structured process controls and identifying issues / root cause before they escalated - the same corrective-action thinking used in OOS management.
- Kept documentation audit-ready throughout every shift. Inspection readiness is not a one-off preparation; it is a daily standard.
- Promoted internally on merit. The promotion reflects consistent performance under sustained operational pressure, not seniority or time-served.

Location & Availability: Based in Leicester. Fully available and committed to relocating for the right pharmaceutical role. Open to roles across the UK.

QUALITY MANAGEMENT SYSTEMS

Foundation of Quality Management Systems

Across my MSc training, BPharm studies, and independent professional development, I have built a structured understanding of Quality Management Systems (QMS) as they apply in pharmaceutical and regulated laboratory environments. This knowledge underpins how I approach every aspect of laboratory and documentation work from daily record-keeping to understanding the regulatory framework that governs batch release and product safety.

GMP & GLP

› Good Manufacturing Practice principles and their application in pharmaceutical production and QC environments

› Good Laboratory Practice: ensuring traceability, integrity, and reproducibility of laboratory data

› Understanding of batch records, deviation reports, and change control procedures

› Audit-readiness as a daily standard: applied directly in Royal Mail Team Lead role

Documentation & Data Integrity

› SOP writing, review, and version control: including a personalised SOP produced during the mammalian cell culture module

›ALCOA+ principles: Attributable, Legible, Contemporaneous, Original, Accurate

› Audit-ready record-keeping across both laboratory and operational environments

› LIMS workflows (self-study in progress) and electronic data management principles

Statistical Process Control (SPC)

ISO Standards

› Understanding of control charts, process capability (Cp, Cpk), and trend monitoring as tools for detecting process drift before OOS events occur

› Applied statistical analysis during MSc HPLC method optimisation : interpreting purity data across five reaction time conditions to identify the optimal parameter set

› Familiarity with OOS and OOT investigation workflows and CAPA frameworks

› ISO 9001: Quality Management System requirements: process approach, risk-based thinking, continual improvement

› ISO/IEC 17025: Competence requirements for testing and calibration laboratories, including method validation and measurement uncertainty

› ICH Q10 Pharmaceutical Quality System: lifecycle approach to quality from development through manufacture

› ICH stability guidelines (Q1A–Q1F): applied directly in the 14-day stability study component of MSc research

QMS Knowledge — Applied Context

QMS Element	Knowledge Base	Applied Evidence
GMP Documentation	Udemy: Quality in the Pharmaceutical Industry	Audit-ready records across Royal Mail reporting cycles; SOP written during MSc cell culture module
GLP & Data Integrity	MSc training + self-study (ALCOA+ principles)	HPLC and FTIR data recorded and processed to GLP standard throughout MSc research project
Statistical Process Control	MSc analytical method optimisation	Purity data analysed across 5 reaction time conditions; optimal parameter identified and justified statistically
ISO 17025	Self-study: laboratory competence framework	Understanding of method validation, measurement uncertainty, and accreditation requirements for testing laboratories
ICH Stability (Q1A)	MSc 14-day stability study	UV absorbance monitoring across 300–600nm; storage condition assessment; degradation trend analysis
OOS & CAPA	GMP coursework + operational application	Root cause identification applied in Royal Mail zero-error performance; OOS thinking embedded throughout MSc lab training

CERTIFICATIONS

Professional Development

Certification	Issuing Body	Date	Status
High Performance Liquid Chromatography (HPLC)	Udemy	March 2025	Completed
Quality in the Pharmaceutical Industry (GMP, GLP, GHP)	Udemy	April 2025	Completed
Control of Substances Hazardous to Health (COSHH)	Alison	March 2026	Completed
Internal Audit Training	Alison	2026 / ongoing	In Progress
LIMS Workflows & Data Integrity Principles	Self-study	Ongoing	In Progress

CAREER OBJECTIVE

Career Objective

I am applying for QC Analyst and Analytical Scientist roles within UK pharmaceutical manufacturing and contract research organisations. My technical background covers the core analytical techniques used in pharmaceutical QC: RP-HPLC, ATR-FTIR, UV-Vis spectroscopy, alongside microbiology skills from both my BPharm and MSc training. I understand GMP and QMS principles, I can write and follow SOPs, and I have demonstrated that I can maintain accurate, auditable records under pressure.

I recognise that I am entering the industry without commercial laboratory experience, and I am not trying to argue otherwise. What I can offer is solid technical grounding, a genuine understanding of why quality matters in this industry, and a track record of performing well when the responsibility is real. If there are skills the role requires that I do not yet have, I will develop them. That is not a vague intention, I am currently studying data integrity principles and LIMS workflows alongside my job search. I am looking for the first opportunity, and I will make the most of it.

Open to Graduate QC Analyst, Analytical Scientist, and Microbiologist roles across the UK.

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